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MILITARY SPECIFICATION

CIRCUIT BREAKERS, NON TRIP-FREE

GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the general requirements for two types of thermal operated, non trip-free circuit breakers used in aircraft propeller feathering and propeller control electric circuits. (See 6.1)

1.2 Classification.- Circuit breakers shall be of the following types:

Type I	Push-Pull, button actuated
Type II	Toggle, lever actuated

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes, Set-Up, Paperboard

FSC 5925

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Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-P-997	Plastic Material; Laminated, Thermosetting, Electrical Insulating; Sheets, Glass Cloth, Silicone Resin
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (for Production Aeronautical and Associated Equipment)
MIL-W-5086	Wire, Electrical, 600-Volt, Copper, Aircraft
MIL-E-5272	Environmental Testing; Aeronauitcal and Associated Equipment, General Specification for
MIL-S-7742	Screw Threads; Standard, Aeronautical
MIL-L-10547	Liners, Case, Waterproof
MIL-P-15037	Plastic Material; Laminated, Theremosetting, Sheets, Glass Cloth, Melamine-Resin
MIL-N-25027	Nut, Self-Locking, 250°F, 550°F and 800°F
MIL-S-27220	Shield, Environmental, Aircraft, Push-Pull Circuit Breaker

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking for U.S. Military Property

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MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

DRAWINGS

Air Force-Navy Aeronautical Standard Drawing

AN3179	Inductor; 30 Volt Laboratory Test
MS20659	Terminal, Lug, Crimp Style, Copper, Uninsulated
MS25036	Terminal, Lug, Crimp Style, Copper, Insulated

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The requirements in the applicable MS standards are applicable as detail requirements of this specification. Whenever the requirements of this specification and the applicable MS standard conflict, the requirements of the MS standard shall govern.

3.2 Qualification.- The circuit breakers furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.3 Materials.- Materials shall conform to applicable specifications and shall be as specified herein. Materials not covered by applicable specifications or not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. Plastic materials which are non-nutrient to fungus shall be used in construction of circuit breaker.

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3.3.1 Metals.- All metals used in construction of circuit breakers shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion. The use of dissimilar metals, especially contacts between brass, copper, or steel and aluminum or magnesium alloys, shall be avoided. Where contact between dissimilar metals is unavoidable, the metals shall be protected against electrolytic corrosion. Corrosion resulting from tests specified herein shall not affect performance of the breaker.

3.3.1.1 Contact Resistance.- Circuit breaker contacts shall maintain low electrical contact resistance when subjected to severe environmental conditions and during normal storage conditions for extended periods of time. The contact resistance, measured as specified in paragraph 4.6.21 shall not increase more than 10 millivolts.

3.3.2 Plastics.- Plastic materials used in the housing, insulator base, and any internal parts exposed to arcing or surface creepage shall conform to Specification MIL-P-997, MIL-P-15037, or MIL-M-14. Other types of plastic materials may be used provided the manufacturer submits satisfactory evidence to the activity responsible for qualification that the materials are suitable for the purpose intended. The plastic materials used shall neither support combustion nor give off noxious gases when subjected to arcs, such as those caused by interrupting heavy short-circuit currents, or explosions of gaseous vapors to which the materials may be subjected in service. Plastic materials with cellulose fillers will not be permitted in parts that may be subjected to arcing or surface creepage.

3.3.3 Protective Treatment.- The use of any protective coatings that will crack, chip, or scale will not be permitted.

3.3.4 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.3.4.1 Standard Parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts, such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are

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replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is not suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3.5 Cleaning.- Prior to final assembly, the circuit breaker shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material. Burrs, sharp edges, and resin flash shall be removed.

3.4 Design and Construction.-

3.4.1 General.- Circuit breakers shall conform to the requirements specified on the applicable MS standards and in accordance with this specification.

3.4.2 Non Trip-Free Operation.- The circuit breakers shall be of such design that the electrical current can be maintained by manually holding the breaker actuator in the closed position (on) until fault current destroys breaker.

3.4.3 Mounting Means.- The circuit breaker shall be provided with a suitable mounting means as shown on the applicable MS standard. If self-locking nuts are used, they shall meet the performance requirements of Specification MIL-N-25027.

3.4.4 Actuator of Push-Pull Circuit Breakers.- The portion of the actuator visible when the circuit breaker is in closed position shall be black and shall expose a white band when in the open or tripped position. The exterior portion of the actuator shall be insulated from all current-carrying parts. The actuator shall not work out to an intermediate position, give a false trip indication, or be removable from the breaker.

3.4.4.1 Manual Circuit Opening.- Push-pull circuit breakers shall be designed to permit manual opening of the circuit by pulling out the actuator to the full tripped position as shown on the applicable MS or AN standard.

3.4.5 Actuator of Toggle Breakers.- The exposed portion of the actuator shall be provided with a nonglaring finish and shall be electrically insulated from all current carrying parts.

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3.4.5.1 Manual Circuit Opening.- Toggle breakers shall be designed to permit manual opening of the circuit by moving actuator to tripped (off) position.

3.4.6 Environmental Protection.- Internal parts of push-pull circuit breakers shall be protected from environmental exposure by installation of actuator shield MS27220. Circuit breakers shall meet the environmental requirements of this specification with the shield removed.

3.4.7 Terminals.- Terminal construction shall be as specified in the applicable MS standard and shall be designed for use with terminal lugs conforming to MS25036 or MS20659.

3.4.7.1 Terminal Hardware.- Lockwashers captive to the terminal screws may be supplied.

3.4.8 Housing.- The circuit breaker mechanism shall be enclosed in a housing securely attached to the insulator base and to the mounting plate when one is used. The housing may be integral with the insulator base.

3.4.9 Ratings.- The ratings of push-pull and toggle circuit breakers shall be as specified on the applicable MS standards.

3.4.10 Trip Indication and Reset.- The circuit breakers shall be so designed that when the breaker contacts open automatically on overload, the actuator shall indicate the operation by moving to the tripped position, as shown on the applicable MS standard. Resetting shall be accomplished in accordance with the applicable MS standard.

3.4.11 Position.- Circuit breakers shall operate satisfactorily when mounted in any position.

3.4.12 Mounting Screw Clearance.- The mounting screws shall be capable of being screwed into the circuit breaker a minimum depth, as shown on the applicable MS standard. The mounting nuts shall be backed or provided with other means to prevent mounting screws of excess length from interfering with the operation of the breaker. Screws of excess length shall not fracture the housing or the explosion-proofing seal.

3.4.13 Creepage and Clearance Distance.- The minimum creepage path and the minimum clearance between current-carrying parts and any part of the circuit breaker other than insulating material, and also between current-carrying parts of opposite polarity, shall be 3/16 inch and 1/8 inch, respectively.

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3.4.14 Tamper-Proof Calibration.- Breakers shall be so sealed that tampering with the calibration is not possible without dismantling the device or breaking a seal.

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.6 Screw Threads.- Screw threads on removable or replaceable threaded parts shall be as specified in MIL-S-7742. Threading on nonmetallic parts is not permitted.

3.7 Performance.- The circuit breakers shall perform satisfactorily when subjected to any natural combination of the following tests specified in Section 4 and there shall be no breakage, malfunction, or evidence of any defect which could impair ability of the breakers to meet the requirements of subsequent tests.

(a) Dielectric Strength, Para 4.6.2 .

(b) Insulation Resistance, Para 4.6.3.

(c) Strength of Actuator, Para 4.6.4.

(d) Strength of Threaded Parts, Para 4.6.5.

(e) Operating Force, Para 4.6.6.

(f) Calibration, Para 4.6.7.

(g) Endurance, Para 4.6.8.

(h) Overload Cycling, Para 4.6.9.

(i) Reclosing, Para 4.6.10.

(j) Operation at Maximum Temp, Para 4.6.11.

(k) Hold-in Capacity, Para 4.6.12.

(l) Rupture Capacity, Para 4.6.13.

(m) Voltage Drop, Para 4.6.14.

(n) Vibration, Para 4.6.15.

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(o) Mechanical Shock, Para 4.6.16.

(p) Acceleration, Para 4.6.17.

(q) Sand and Dust, Para 4.6.18.

(r) Corrosion, Para 4.6.19.

(s) Humidity, Para 4.6.20.

(t) Contact Resistance, Para 4.6.21.

(u) Explosion Proof, Para 4.6.22.

(v) Hold-in Rupture, Para 4.6.23.

3.8 Markings.- All circuit breakers shall have the MS dash number permanently and legibly marked on the terminal end of the breaker. Other markings shall be as shown on applicable MS Drawings.

3.8.1 Push-Pull Actuators.- The exposed end of the push-pull actuator shall be marked with a white or silver, raised or depressed, number indicating the applicable MS dash number of the breaker.

3.8.2 Toggle Actuators.- The tip of actuators of toggle breakers shall be marked with a raised or depressed number indicating the applicable MS dash number.

3.8.3 Line Terminals.- The line terminals shall be clearly and permanently marked "LINE" adjacent to the terminals, as shown on the applicable MS standard.

3.9 Identification of Product.- Each circuit breaker shall be permanently and legibly marked for identification in accordance with MIL-STD-130 as follows:

MS (applicable drawing number complete with dash number)
Manufacturer's part number
Manufacturer's name or trade-mark and coded date 1/

1/Date code shall consist of "MFD" followed by a four digit number; in which the first two digits will indicate the month, and the last two digits will indicate the year, e.g., "MFD-0564" would indicate a unit manufactured in May 1964.

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3.10 Workmanship.- The circuit breaker, including all parts and accessories shall be constructed and finished in a careful and workmanlike manner in accordance with good design and sound practice. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges (See 4.5.1).

4. QUALITY ASSURANCE

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests.- The inspection and testing of circuit breakers shall be classified as follows:

(a) Qualification Tests, Para 4.4.

(b) Quality Conformance Tests, Para 4.5.

4.3 Test Conditions.-

4.3.1 Atmospheric Conditions.- Unless otherwise specified, all tests shall be conducted in still air at a temperature of 25° + 10°, -5°C with a relative humidity no greater than 80 percent. The atmospheric pressure may be used in lieu of sea level pressure.

4.3.2 Electric Currents.- Unless otherwise specified, test currents shall be maintained at a constant value by automatic or manual control and tests may be conducted using alternating current at commercial frequencies.

4.3.3 Test Equipment.- Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspections. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government. Ammeter accuracies shall be within .5 of 1 percent.

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4.3.3.1 Test Cables.- Test cables in accordance with Table I shall be used for all tests that require circuit breakers to carry current except tests requiring currents of fault magnitudes in the rupture and explosion-proof tests. Wires and terminals shall conform to MIL-W-5086.

TABLE I

(a) CIRCUIT-BREAKER CAPACITY (AMPERES)	WIRE SIZE (AN DESIG.)	UNINSULATED TERMINAL LUG (MS P/N)	INSULATED TERMINAL LUG (MS P/N)
LESS THAN 6	18	MS20659-2	MS25036-3
7 TO 10 INCL	16	MS20659-4	MS25036-8
11 TO 15 INCL	14	MS20659-4	MS25036-8
16 TO 20 INCL	12	MS20659-5	MS25036-12
21 TO 25 INCL	10	MS20659-5	MS25036-12
26 TO 40 INCL	8	MS20659-7	MS25036-15
41 TO 50 INCL	6	MS20659-30	MS25036-19
51 TO 60 INCL	6	MS20659-31	MS25036-21

4.3.4 Calibration.- Calibration tests of circuit breakers shall be conducted at a temperature specified in the applicable MS standard with the current maintained at a constant value. Test samples with leads and terminals attached in accordance with Figure I shall be maintained at the specified ambient temperature for 1 hour prior to application of the specified current during all Calibration Tests. Test circuit may be in accordance with Figure 1.

4.3.4.1 Production Calibration Tests.- For Calibration Tests performed during production only (not for Qualification or Acceptance Tests), it will be permissible to use test methods so designed as to expedite calibration testing, provided the vendor can demonstrate to the satisfaction of the Inspector that the results are comparable to the Calibration Tests specified herein.

4.3.5 Rupture Capacity.- For the Rupture-capacity tests specified in 4.6.13 the circuit breaker shall be so connected to the power source that currents specified on the applicable MS standard are provided at the circuit-breaker terminals. The test circuit of Figure 2 may be used. The open-circuit voltage before application of the rupture current shall be as specified on the applicable MS standard. The open-circuit recovery voltage shall be the value specified on the applicable MS standard. Oscillographic records of current, voltage and time shall be obtained.

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4.4 Qualification Tests.- Qualification test for circuit breakers of each rating shall consist of all the tests specified in 4.6. Each test sample shall be subjected to those tests listed in the order shown in Table III. Subsequent modification of the approved product shall not be permitted without approval of the qualifying activity.

4.4.1 Sampling Instructions.- Qualification test samples shall consist of 24 circuit breakers of each rating for which approval is sought. Test samples shall be representative of the manufacturer's normal production and shall be identified by the manufacturer's part number and test sample number.

4.4.1.1 Test Samples.- Test samples shall be submitted for qualification testing as designated by the activity responsible for qualification. Certified test data covering nondestructive tests for the type circuit breaker for which approval is sought shall be submitted to the activity responsible for qualification. This report shall be submitted prior to qualification testing. Samples shall be plainly identified by securely attached durable tags marked with the following information: (See 6.3)

Sample for qualification test
Circuit breaker
Name of manufacturer
Circuit Breaker, designation military standard and dash number
Submitted by (name) (date) for qualification tests in accordance with the requirements of MIL-C-7079 under authorization (reference authorizing letter)

4.4.1.2 Test Sample Modification.- All test samples, of the same rating, shall be identical, in all respects, to each other and modification of one or more samples in any way shall not be permitted without approval of the qualifying activity.

4.5 Quality Conformance Tests.- Quality conformance tests shall consist of:

(a) Individual Tests, Para 4.5.1.

(b) Sampling Tests, Para 4.5.2.

4.5.1 Individual Tests.- Each circuit breaker shall be subjected to examination of product as specified in 4.6.1 and calibration at 200 percent rated current at 25°C as specified on applicable MS standard. Breakers failing three tests shall be rejected and may be subsequently retested.

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4.5.2 Sampling Tests.- Statistical sampling and inspection shall be in accordance with MIL-STD-105 and consist of those tests specified in Table II. Inspection Level I shall be used for determining sample size. Acceptable quality levels (AQL) shall be as specified in Table II. Major and minor defects shall be as specified in MIL-STD-105.

4.5.2.1 Lot.- The lot definitions, formation, and size shall be in accordance with MIL-STD-105.

TABLE II

TEST	TEST PARA	AQL Percent Defective	
		Major	Minor
Examination of Product		0	4.0
Dielectric Strength		0	4.0
Insulation Resistance		0	4.0
Operating Force		2.5	4.0
Overload Cycling		1.0	4.0
Hold-in Capacity		1.0	4.0
Calibration		1.0	4.0

4.6 Test Methods.-

4.6.1 Examination of Product.- Circuit breakers shall be inspected to verify that the materials, design, construction, weight, physical dimensions, marking, and workmanship conform to the applicable requirements. (See 3.3, 3.4, 3.6, 3.8, 3.9, and 3.10.)

4.6.2 Dielectric Strength.- The circuit breaker shall withstand 1,000 volts (rms) plus twice maximum rated voltage ac at commercial frequency (1,500 volts minimum) and shall show no evidence of breakdown, flashover, or current flow in excess 1.0 milliamperes under the following conditions:

(a) Line and load terminals with the circuit breakers in the off, or tripped, position.

(b) Terminals and parts normally grounded (such as frame, shell, mounting plate, etc.) with the circuit breakers in closed, and on position.

The potential shall be applied at a maximum rate of increase of 250 volts per second, until the test potential is reached, and shall be maintained for 1 minute during Qualification Tests. During acceptance tests, a potential equal to 120 percent of the above values may be applied for a duration of 5 seconds.

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TABLE III - QUALIFICATION TEST SCHEDULE

PARA.	QUAL TEST	PARA	TEST SAMPLE NUMBERS														
			1 2	3	4	5	6 7	8 9	9 10	11	12 13	14 15	16 17	15 18 20	21 22	23 24
4.6.1	Exam of Product	4.6.1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3.7	Dielectric Strength	4.6.2	7	2	2	2	2	4	3	3	3	3	2		2	
3.7	Insulation Resistance	4.6.3	2	3	3											
3.7	Strength of Actuator	4.6.4	3	4												
3.7	Strength of Threaded Parts	4.6.5	4													
3.7	Operating Force	4.6.6	6	5	4	3										
3.7	Calibration	4.6.7														2
3.7	Endurance	4.6.8	5													
3.7	Overload Cycling	4.6.9		6												
3.7	Reclosing	4.6.10			5											
3.7	Operation at Max Temp	4.6.11				4										
3.7	Hold-in Capacity	4.6.12				5										
3.7	Rupture Capacity	4.6.13					3									
3.7	Voltage Drop	4.6.14						2						2		
3.7	Vibration	4.6.15						3								
3.7	Mechanical Shock	4.6.16							2							
3.7	Acceleration	4.6.17								2						
3.7	Sand and Dust	4.6.18									2					
3.7	Corrosion	4.6.19										2				
3.7	Humidity	4.6.20											3			
3.3.1.1	Contact Resistance	4.6.21												3		
3.7	Explosion-Proof	4.6.22													3	
3.7	Hold-in Rupture	4.6.23														3

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4.6.2.1 Where the Dielectric-strength test is called for following another test, the Dielectric-test voltages shall be reduced to 75 percent of the value specified.

4.6.3 Insulation Resistance.- The insulation resistance shall be measured between mutually insulated metal parts, by means of a megohm bridge or other approved method, at a d-c potential of 500 volts for Qualification tests, and at 500 to 1,000 volts for Acceptance tests. Unless otherwise specified, the insulation resistance of circuit breakers shall be not less than 100 megohms.

4.6.4 Strength of Actuator.- There shall be no malfunction, breakage, or evidence of damage during the following tests.

4.6.4.1 Strength of Actuating Lever Pivot and Lever Stop.- The lever pivot and stop shall be subjected to a 25-pound load applied for 1 minute to the tip of the actuating lever as follows:

(a) Perpendicular to the lever axis and parallel to the line of lever travel at each end position of the lever.

(b) Same as condition (a), but perpendicular to the line of travel at each position of the lever.

(c) Coaxial with the lever axis toward the lever pivot throughout the entire range of travel of the lever.

(d) Coaxial with the lever axis away from the lever pivot throughout the entire range of travel of the lever.

4.6.4.2 Strength of Push Button.- A 25-pound force shall be applied for 1 minute in both directions, along the line of push-button travel. With the push button in the fully extended position, a force of 25 pounds shall be applied at the extremity for 1 minute in two mutually perpendicular directions, each normal to the line of push-button travel.

4.6.5 Strength of Threaded Parts.-

4.6.5.1 Strength of Terminals.- The tensile load specified in Table IV shall be applied for 1 minute to each terminal, perpendicular to the mounting plate of the circuit breaker, successively. The torque specified in Table IV shall then be applied to the screw head about the thread axis for 1 minute without damage to the terminals.

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TABLE IV
Strength of Threaded Parts

Terminals		
Screw Threads	Tensile Load (lb)	Torque (lb. in.)
8-32 NC-2A	25	15
5/16-24 UNF-2A	70	80
Mounting Means		
Screw Threads	Axial Loads (lb)	Torque (lb. in.)
6-32 NC-3B	30	10
8-32 NC-3B	35	20
15/32-32 NS-2A	--	50

4.6.5.2 Strength of Mounting Means.- The axial load specified in Table IV shall be applied to the mounting means for a period of 1 minute, using hardened steel screws. The torque specified in Table IV shall then be applied to the screw head about the thread axis for 1 minute without damage to the mounting means. For single-hole-mounting circuit breakers, this test shall be conducted with the mounting nut assembled on the threaded bushing of the breaker.

4.6.6 Operating Force.- The force necessary for operation of the circuit breaker shall be determined. For toggle breakers, the force shall be applied at the ball tip of the actuator, and in the direction of operation. For push-pull breakers, the force shall be applied parallel to the line of travel of the actuator. The forces required for operation of the breakers either during or as a result of the tests shall not exceed the values specified on the applicable MS standard.

4.6.7 Calibration.- Circuit breakers shall be subjected to Calibration tests given in the following paragraphs.

4.6.7.1 Minimum Limit of Ultimate Trip.- The circuit breaker shall be subjected to the minimum limit of ultimate trip current for the time specified on the applicable MS standard. For Qualification tests only, the temperature rise of the breaker terminals shall be obtained by the use of a suitable thermocouple. The breaker shall not trip, and, unless otherwise specified on the applicable MS standard, the temperature rise

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measured at the terminals shall not exceed 75°C. The thermocouple shall be attached to terminals outside but adjacent to the breaker case.

4.6.7.2 Maximum Limit of Ultimate Trip.- The circuit breaker shall be subjected to the maximum limit of ultimate trip current specified on the applicable MS standard and shall trip within the time limits specified.

4.6.7.3 Overload Calibration.- The circuit breaker shall be subjected to the overload calibration values shown on the applicable MS standard, and operation shall be within the limits specified.

4.6.7.4 Ambient Effect on Calibration.- The circuit breaker shall be tested at ambient temperatures and loads specified on the applicable MS standard and shall operate within the time limits specified.

4.6.8 Endurance.- The circuit breaker shall be subjected to the number of cycles of make-and-break operations specified in the applicable MS standard and shall make-and-break the specified current throughout the cycling period. Operating cycle is defined as the mechanical opening and closing of the breaker. Operation shall be performed at a minimum rate of 2 cpm or at a faster rate, and the ratio of "on" time to "off" time shall be less than 1 to 5. The mechanical operation shall simulate manual operation of the breaker, including overtravel. Upon completion of the test, the breakers shall calibrate with 90 percent of the minimum ultimate trip current and 110 percent of the maximum ultimate trip current.

4.6.8.1 Alternating Current.- During the alternating-current Endurance tests the voltage and frequency shall be as specified on the applicable MS standard.

4.6.8.1.1 Inductive Load.- The power factor during this test shall be between 0.6 and 0.7 lag unless otherwise specified by the applicable MS standard. The load shall be the rating of the circuit breaker unless otherwise specified by the applicable MS standard.

4.6.8.1.2 Resistive Load.- Resistive operation shall be accomplished at rated load with a power factor between 0.9 and unity.

4.6.8.2 Direct Current.- During the direct-current Endurance tests the voltage shall be within the limits specified on the applicable MS standard.

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4.6.8.2.1 Inductive Load.- The inductive cycles shall be performed using inductors as specified in Drawing AN3179. For ratings below 10 amperes the load shall be rated current. For ratings 10 amp and above the load shall be 10 amp.

4.6.8.2.2 Resistive Load.- Resistive operations shall be accomplished as rated load.

4.6.8.3 Mechanical Cycling.- The circuit breaker shall be subjected to the number of cycles of opening and closing specified on the applicable MS standard. The rate of cycling shall not exceed 60 operations per minute.

4.6.9 Overload Cycling.- At 200-percent-rated-current resistive load, 100 cycles of manual make and automatic break shall be applied. This test shall be run at 30 ± 2 volts dc and at a manual cycling rate of 2 to 3 minutes per operation. In the absence of a d-c rating, the test shall be conducted at its rated a-c voltage. Failure to trip automatically throughout 100 consecutive cycles shall constitute failure. Following this test, the breaker shall be subjected to minimum and maximum ultimate trip tests as specified in 4.6.7.1 and 4.6.7.2.

4.6.10 Reclosing.- Manual-reset circuit breakers shall remain open after being tripped automatically and subjected to an ambient temperature of -60° ± 5°C for 1 hour. Continuity shall be checked during this low-temperature exposure. This test may be waived if the design of the breaker precludes automatic closing.

4.6.11 Operation at Maximum Temperature.- The circuit breakers shall be tested to determine, that after tripping at an ambient temperature of 57°C, they can be reset within 30 seconds at 57°C and remain closed for 10 minutes at 100 percent rated load at this same temperature.

4.6.12 Hold-in Capacity.- The hold-in capacity test shall be conducted in any manner satisfactory to the inspector. The circuit breakers shall not fuse closed when held closed against a 400 percent rated current for 10 seconds and 250 percent rated current for 90 seconds. The operations shall be in the order specified and with a minimum of 15 minutes between each operation. Following this test, the circuit breaker shall pass overload calibration test, specified in 4.6.7.3, at 200% rated current.

4.6.13 Rupture Capacity.- The circuit breakers shall be subjected to five rupture currents as specified on the applicable

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MS standard, with test conditions specified in 4.3.6 and shall open with the rupture currents and voltages specified. After each rupture, the open-circuit voltage specified by the applicable standard shall be maintained across the breaker for a minimum of 5 seconds. There shall be sufficient time to permit proper cooling and reset between each cycle of operation. Under no circumstances shall a rupture test be repeated within 5 minutes of the previous test. Following the last operation of each test, the breaker at room ambient shall be subjected to the Dielectric strength test (4.6.2) and to calibration at 200 percent of rated current. The breaker tripping time shall be within 80 percent of the lower limit and 120 percent of the upper limit specified in 4.6.7.3 and there shall be no dielectric breakdown. Each test sample shall be subjected to three rupture currents at an ambient pressure altitude of 65,000 feet and two rupture currents at ambient room pressure. Where more than one rupture current value is specified on the MS standard, an additional breaker for each value shall be tested.

4.6.14 Voltage Drop.- The voltage drop of the circuit breaker from terminal to terminal of one pole shall be measured while subjected to rated current, and the voltage drop, as measured from terminal to terminal, shall not exceed the value specified by the applicable MS standard. This test shall be performed after the breaker has been carrying its rated current for at least 30 minutes. The power loss shall be determined. In no case shall the power dissipation exceed 15 watts.

4.6.15 Vibration.- While carrying rated load, the circuit breaker shall be subjected to the vibration test, Procedure XII of MIL-E-5272. For Vibration tests, run at 71°C, the breaker shall carry 90 percent of minimum ultimate trip current specified by the applicable MS standard for this temperature. The breaker contacts shall remain closed. An oscilloscope, oscillograph, or chronoscope shall be utilized to determine opening or closing of contacts. Following the test, the breaker shall show no evidence of damage caused by vibration and shall meet the requirements of the 200-percent-overload test specified in 4.6.7.3.

4.6.16 Mechanical Shock.- Circuit breakers shall be subjected to the Shock test, Procedure I of MIL-E-5272, in each of the three principal axes in turn. The sample being tested shall be secured to sufficient mass, and the carriage dropped through sufficient distance, to result in a deceleration of 25g with a duration of 11 ± 1 milliseconds when decelerated by resilient impact. Three separate shocks shall be applied to each of the three principal axes with the breaker contacts in

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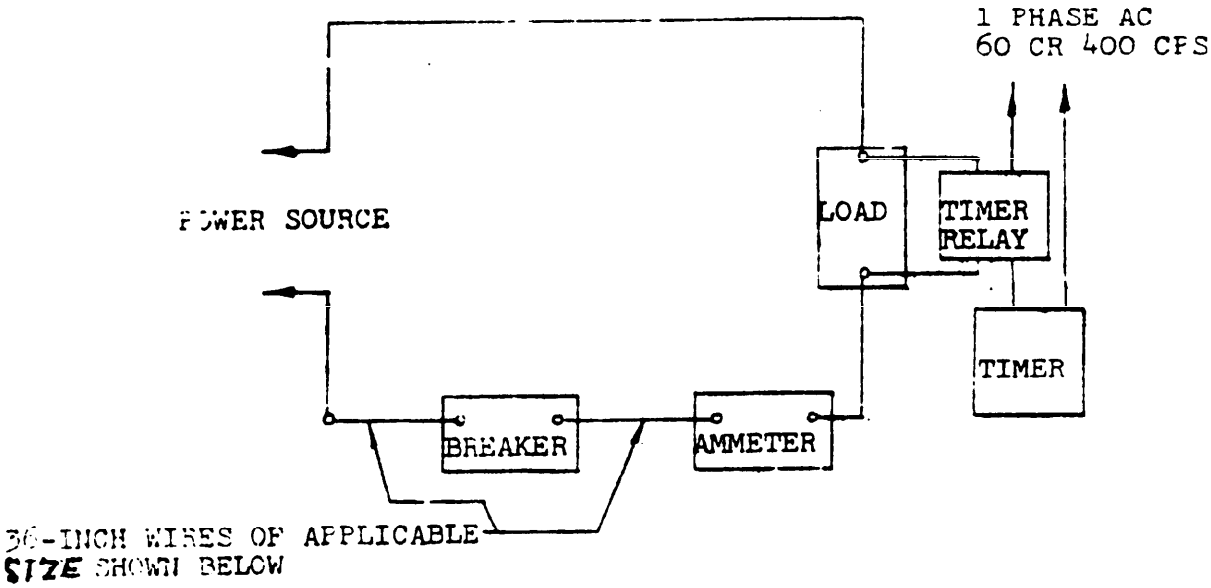
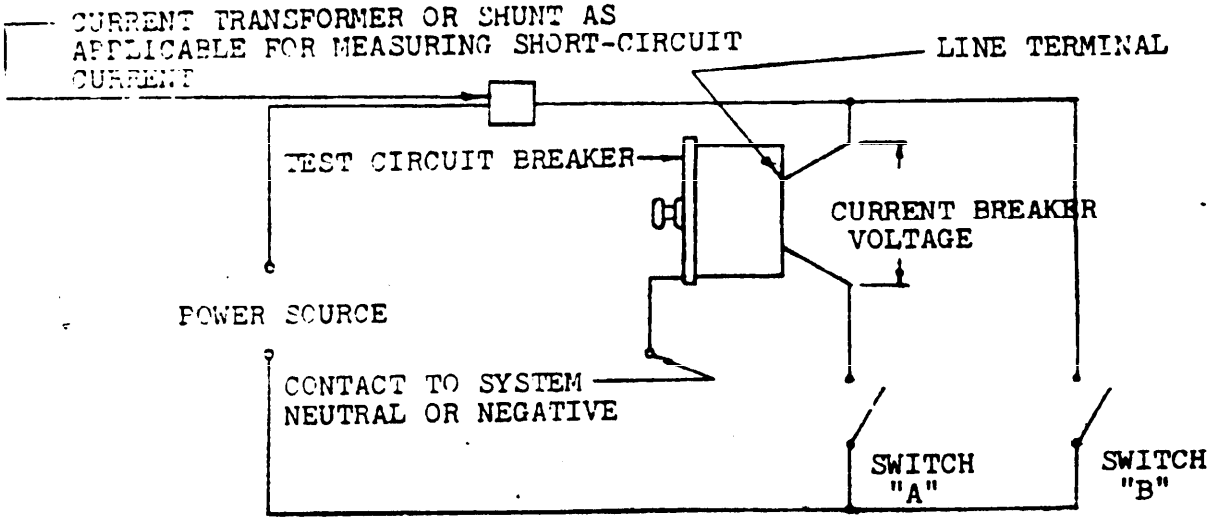


FIGURE 1. CALIBRATION TEST CIRCUIT



(a) A-C or D-C SHORT-CIRCUIT RUPTURE TEST

OPEN TEST CIRCUIT BREAKER AND SWITCH "A".
CLOSE SWITCH "B" AND DETERMINE IF CURRENT CAPACITY OF CIRCUIT IS IN ACCORDANCE WITH APPLICABLE STANDARD OF TEST BREAKER.
OPEN SWITCH "B".
CLOSE TEST CIRCUIT BREAKER.
CLOSE SWITCH "A".

(b) RUPTURE CURRENT SHALL BE MEASURED BY A CURRENT TRANSFORMER OR SHUNT AS INDICATED AND SUITABLE RECORDING OSCILLOGRAPH. THE VOLTAGE ACROSS THE BREAKER SHALL BE RECORDED SIMULTANEOUSLY WITH RUPTURE CURRENT AND AT THE POINT INDICATED.

FIGURE 2. RUPTURE-TEST CIRCUIT

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the closed position, and three separate shocks shall be applied to each of the axes with the breaker contacts in the open position. The breaker shall be carrying rated current during the test and for 30 minutes prior to it. A chronoscope, an oscilloscope, or other device, capable of detecting momentary opening or closing periods not exceeding 1/2 millisecond duration, shall be used to determine that the breaker contacts in the closed position remain closed, and circuit-breaker contacts in the open position remain open. Following the test, the breaker shall be subjected to the 200-percent-overload test specified in 4.6.7.3.

4.6.17 Acceleration.- The circuit breaker shall be subjected to a continuous acceleration of 10g. The breaker shall be mounted in the position most likely to cause malfunctioning and shall be accelerated at 10g for a minimum of 1 minute. The breaker shall be carrying 100 percent rated current during the test and for 30 minutes prior to it. Test shall be repeated with the breaker contacts in the open position. An oscilloscope, oscillograph, or a chronoscope shall be used to determine the ability of the breaker contacts to remain in the proper position. There shall be no opening or closing of contacts, and there shall be no evidence of damage caused by acceleration. Following the test, the breaker shall be subjected to the 200-percent-overload test specified in 4.5.7.3.

4.6.18 Sand and Dust.- While in the "on" position and mounted on a dummy panel, the circuit breaker shall be subjected to sand and dust in accordance with the Sand and Dust tests, Method 110, Test Condition A of MIL-STD-202 with no evidence of mechanical or electrical failure. The breaker shall meet the requirements for calibration at room ambient and 200 percent of rated load as specified in 4.6.7.3.

4.6.19 Corrosion.- The circuit breaker in the "on" position shall be subjected to the Salt Spray test, Method 101, Test Condition B of MIL-STD-202, with a 5 percent salt solution. Within 10 minutes after the test, the breakers shall be washed for 5 minutes under running water not warmer than 37.8°C accompanied by a slight brushing, and dried for 6 hours in a forced-draft oven at approximately 57°C. The breaker shall be subjected to and shall satisfactorily clear an overload equal to 200 percent of rated current and shall then meet the requirements for calibration at room ambient with 200 percent of rated currents, except that the tripping time shall be within 80 percent of the lower limit and 120 percent of the upper limit specified.

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4.6.20 Humidity.- The circuit breaker in the "on" position shall be subjected to the Humidity test, Procedure I, of MIL-E-5272. Within 1 hour after removal from the test chamber and without forced drying, the breaker shall be subjected to 500 percent of rated load. Failure to trip within 10 seconds shall result in disqualification. Following the 500-percent-rated-load test, the circuit breaker shall trip free when closed at 300 percent of rated load and shall then meet the requirement of calibration at 200 percent of rated load, except that the tripping time shall be within 90 percent of the lower limit and 110 percent of the upper limit specified therein. The circuit breaker shall show no evidence of breaking, cracking, spalling, excessive corrosion, or loosening of terminals. Following the Humidity test the breaker shall meet the requirements for dielectric strength specified in 4.5.2.

4.6.21 Contact Resistance.- Three circuit breakers in the "on" position, with no electrical load applied, shall be subjected to the Humidity test, Procedure I, of MIL-E-5272. Prior to subjection to the Humidity test, the initial contact resistance of each circuit breaker shall be determined as follows: The circuit breaker contacts shall be manually operated to successively interrupt and make a test circuit having a direct current resistive load of 200 milliamps at 26 volts plus or minus 2 volts. The contact resistance (specified in terms of millivolt drop) shall be computed from the average of ten consecutive measurements of which each is taken at the completion of each successive contact closure. All measurements shall be made across the circuit breaker external electrical terminals, by using identical measurement techniques. The breaker contacts shall then remain closed (undisturbed) until the final ten measurements are made at the completion of the test. Within twenty-four hours after removal of the sample breakers from the test chamber, a single measurement for each breaker shall be made. The breakers shall then be stored for a period of sixty days in an area free from chemicals which give off vapors known to be reactive with metals, at a minimum temperature of 20°C and a relative humidity of not less than 40%. A single measurement for each breaker shall be made at the completion of 15, 30, 45 and 60 days of the storage period. The final contact resistance, at the completion of the sixty day period, shall be determined by the same method described for computing initial contact resistance and shall not exceed the value specified in paragraph 3.3.1.1. Following the test, the breaker shall meet the voltage drop test at rated current as specified in paragraph 4.5.19.

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4.6.22 Explosion-Proof.- The circuit breaker shall be subjected to Procedure I of the Explosion-Proof tests of MIL-E-5272. Ignition of the explosive outside the breaker shall constitute failure. Test method shall be the same as for rupture capacity 4.6.13 with exception that all ruptures shall be performed with breaker contained in explosion chamber at room ambient pressure. Breakers shall pass dielectric strength and calibration tests as required in 4.6.13 at completion of rupture tests.

4.6.23 Hold-In Rupture.- A circuit breaker, not previously tested, shall be closed into and held to complete a maximum rupture current as specified on applicable MS standard. The breaker shall not ignite the explosive mixture and need not be operable after being held closed. Test conditions shall be the same as for explosion tests, 4.6.22.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging (See 6.2).-

5.1.1 Level A.-

5.1.1.1 Cleaning.- Unless otherwise specified, circuit breakers shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying.- None required.

5.1.1.3 Preservative Compounds.- None required.

5.1.1.4 Unit Packaging.- Unless otherwise specified, circuit breakers shall be individually packaged in accordance with submethod IIc of MIL-P-116. Wrapping, cushioning or dunnage materials shall be applied to meet the general requirements of the method of preservation of MIL-P-116 except that humidity indicator may be omitted. The cushioned item shall be sealed within a close fitting bag fabricated in accordance with MIL-B-117, Class E, using heat seal closure. The bagged item shall be additionally inserted into a close fitting container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate Packaging.- Circuit breakers packaged as specified in paragraph 5.1.1.4 shall be placed into cartons conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be of identical size and shape and shall contain identical numbers of identical items. Gross weight of the intermediate package shall not exceed ten lbs.

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5.1.2 Level C.- Circuit breakers shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. The level may conform to supplier's commercial practice for retail distribution when such meets the requirements of this level.

5.2 Packing.-

5.2.1 Level A.- The packaged circuit breakers shall be packed in fiberboard containers conforming to PPP-B-636, Class 2. Box closures shall be as specified for weather-resistant grade boxes in the appendix to PPP-B-636. The center seams, ends, edges and manufacturer's joints shall be sealed with waterproof tape 3 inches wide conforming to PPP-T-60. Banding, (reinforcement requirements) excluding metal, is required and shall be applied in accordance with the provisions outlined in the appendix to PPP-B-636.

5.2.2 Level B.- The packaged circuit breakers shall be packed in fiberboard containers conforming to PPP-B-636, Class 1. Box closures shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C.- Circuit breakers packaged as specified shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. The level in general will conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.4 General.- Exterior containers shall be uniform in size and shape, shall be of minimum cube and tare consistent with the protection required and shall contain identical quantities of identical items insofar as practical.

5.3 Marking.- In addition to any special markings required by contract or purchase order, unit packages, intermediate and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use.- The circuit breakers, covered by this specification, are intended for use as overload and short circuit protection of electrical circuits utilized in aircraft systems conforming to MIL-STD-704 and of other circuits requiring non-trip-free breakers conforming to requirements of this specification.

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6.2 Ordering Data.- Procurement documents should specify the following:

(a) MS part number, including appropriate dash number, and MS title.

(b) Selection of applicable levels of preservation and packaging, and packing.

6.3 Provisions for Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening data, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is classed to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to Commanding Officer, Air Force Logistics Command, Wright-Patterson Air Force Base, Ohio 45433, the activity responsible for qualification.

6.4 Definitions.-

6.4.1 Ultimate Trip Current.- Ultimate trip current is the smallest value of current that will cause tripping of the circuit breaker under a given set of ambient conditions.

6.4.1.1 Ultimate Trip Limits.- The specified limits for ultimate trip current are maximum ultimate trip current and minimum ultimate trip current. At the maximum specified ultimate trip current the breaker will open within the specified time, and at the minimum specified ultimate trip current the breaker will not open.

6.4.2 Line Terminal.- The terminal attached to the isolated stationary contact, with the circuit breaker in the open or tripped position, is considered as the line terminal. When both contacts of a circuit are isolated, only one terminal shall be designated the line terminal.

6.4.3 Push-Pull Circuit Breaker.- Push-pull circuit breakers are those which may be manually actuated by an actuator moving in a direction perpendicular to the plane of the mounting plate.

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6.4.4 Toggle Circuit Breaker.- Toggle circuit breakers are those circuit breakers which may be manually actuated by an actuator which pivots about a point on or within the breaker.

6.4.5 MS Dash Number.- Detailed part number designated on MS standard to indicate electrical current rating of breaker.

6.4.6 Nontrip-Free.- A nontrip-free circuit breaker is a breaker so designed that the circuit can be maintained closed when carrying overload current that would automatically trip the breaker to the open position.

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